MIVA
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Course Name

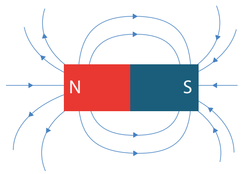
General Practical Physics II

Topic: Magnetic Field and Magnitude of the Magnetic Permeability of Free Space

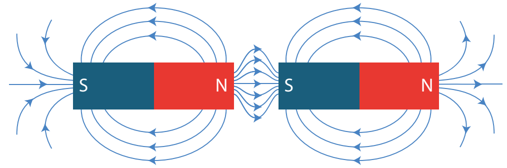
Week 11

IN SUMMARY

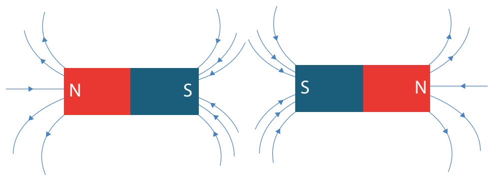
This week, I learned about magnetic fields, their properties, and their applications. I learned that magnetic fields are regions in space where magnetic forces are at play. They can be visualised with field lines, which flow from the north pole of a magnet to its south pole. Magnetic fields are found everywhere, from the Earth's magnetic field to the magnetic fields used in modern technology.



MAGNETIC FIELD OF A BAR MAGNET



ATTRACTION BETWEEN OPPOSITE POLES



REPUTATION BETWEEN LIKE POLES

BULLET POINT SUMMARY

- Magnetic fields are regions in space where magnetic forces are at play.
- Magnetic fields are represented by field lines, which flow from the north pole of a magnet to its south pole.
- The Earth has a magnetic field, which is what causes compass needles to point north and protects us from the harsh solar wind.
- Magnetic fields play a vital role in modern technology, from MRI machines to hard drives.
- Magnetic fields also have practical applications in electric power generation, electric motors, magnetic particle imaging, magnetic sensors, particle accelerators, magnetic resonance spectroscopy, magnetic locks, magnetic separation, space exploration, magnetic bearings, magnetic cooling, Maglev trains, and nuclear magnetic resonance.
- The magnitude of the magnetic permeability of free space is $4\pi \times 10^{-7}$ H/m.

CASE STUDY

Title: Using Magnetic Fields to Improve the Efficiency of Electric Motors

Introduction

Electric motors are used in a wide range of applications, from powering our homes and appliances to driving our vehicles. The efficiency of an electric motor is an important factor in determining its energy consumption and environmental impact.

One way to improve the efficiency of electric motors is to use magnetic fields more effectively. This can be done by using stronger magnets and by designing the motor in a way that maximises the magnetic field strength throughout the motor.

Objective

The objective of this case study is to investigate the use of magnetic fields to improve the efficiency of electric motors.

Methodology

Researchers at the Massachusetts Institute of Technology (MIT) have developed a new type of electric motor that uses a combination of high-strength magnets and a novel motor design

Key references:
Shukla, R. K., & Srivastava, A. (2006). Practical Physics. New Age International (P) Limited, Publishers.
Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.

CASE STUDY

to achieve record-breaking efficiency. The new motor is up to 50% more efficient than traditional electric motors.

The researchers used a variety of techniques to improve the efficiency of the motor, including:

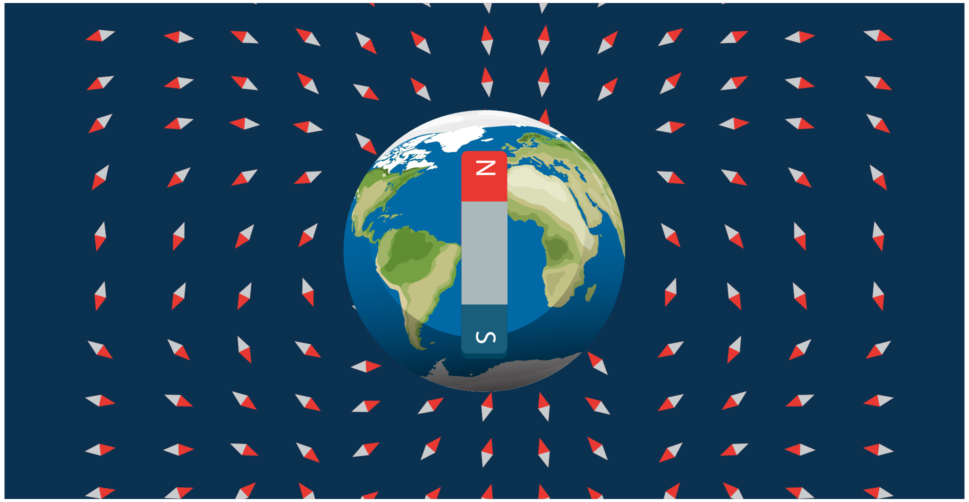
- Using stronger magnets: The new motor uses magnets that are up to 10 times stronger than the magnets used in traditional electric motors.
- Designing a more efficient motor core: The motor core is the part of the motor that contains the magnets and the windings. The researchers designed a new motor core that maximises the magnetic field strength throughout the motor.
- Reducing friction: The researchers reduced friction in the motor by using new materials and lubricants.

Results

The new electric motor developed by the MIT researchers is up to 50% more efficient than traditional electric motors. This means that the new motor can produce the same amount of power using less energy. The new motor is also smaller and lighter than traditional electric motors. This makes it ideal for use in applications where space and weight are at a premium, such as electric vehicles and aircraft.

QUESTIONS TO PONDER

- **What are some other ways that magnetic fields can be used to improve the efficiency of technologies?**
- **What are some of the challenges to developing new and innovative applications for magnetic fields?**



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Ability to define magnetic fields and explain their properties
Ability to discuss the applications of magnetic fields in modern technology
Proficiency in communicating the results of magnetic field experiments in a clear and concise manner.

PERSONAL REFLECTION

I learned a lot about the fundamental properties of magnetic fields and their many applications in the world around us. I am particularly interested in the potential of magnetic fields to improve the efficiency of technologies and to help us develop new and innovative solutions to global challenges.

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